

微积分I-2-A参考答案及评分标准

(2022—2023学年第2 学期) A 卷

一. 填空题 (每题3分, 共18分)

1. $\pm \frac{\sqrt{3}}{2}$. 2. $(x-1) + 3(y-2) + 5(z-2) = 0$ 或 $x + 3y + 5z = 17$.
3. $\frac{1}{8}$. 4. 10π . 5. $(\frac{2}{3}, \frac{4}{3}]$. 6. $-\frac{2}{9\pi}$

二 (共8分). 两边对 x 求偏导: $1 + z'_x = yf' \cdot (2x - 2zz'_x)$, $z'_x = \frac{2xyf'-1}{1+2yzf'}$. 4分

两边对 y 求偏导: $z'_y = f + yf' \cdot (-2zz'_y)$, $z'_y = \frac{f}{1+2yzf'}$. 6分

$z = yf - x$, $zz'_x + yz'_y = \frac{2xyzf'-z+yf}{1+2yzf'} = \frac{2xyzf'+x}{1+2yzf'} = x$. 8分

(法二: 令 $F(x, y, z) = x + z - yf(x^2 - z^2)$, 参照给分)

三 (共10分).

(1) 连续性: $|\frac{x^3+y^3}{\sqrt{x^4+y^4}}| \leq |x| + |y| \rightarrow 0$, $\lim_{\rho \rightarrow 0} \frac{x^3+y^3}{\sqrt{x^4+y^4}} = 0$. 故连续. 2分

可微性: $f'_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0)-f(0,0)}{x} = 1$, $f'_y(0,0) = \lim_{y \rightarrow 0} \frac{f(0,y)-f(0,0)}{y} = 1$. 4分

$\lim_{\rho \rightarrow 0} [f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y] / \rho = \lim_{\rho \rightarrow 0} [\frac{x^3+y^3}{\sqrt{x^4+y^4}} - (x+y)] / \rho$.

当 $y = \pm x$ 时极限不同(分别为 $1 - \sqrt{2}, 0$). 故原极限不存在, 从而不可微. 7分

(2) $\frac{\partial f}{\partial t}|_{(0,0)} = \lim_{\rho \rightarrow 0} \frac{f(t,t)-f(0,0)}{\rho} = \lim_{\rho \rightarrow 0} \frac{f(t,t)}{\sqrt{2}t} = 1$. 10分

四 (共8分). 椭球面上的点 (x, y, z) 到平面的距离为 $d(x, y, z) = \frac{|x+y+z-4|}{\sqrt{3}}$.

目标函数: $f(x, y, z) = (x + y + z - 4)^2$

条件: $g(x, y, z) = x^2 + y^2 + 4z^2 - 1 = 0$

故拉格朗日函数: $L(x, y, z, \lambda) = f(x, y, z) + \lambda g(x, y, z)$. 2分

$$\begin{cases} L'_x = 2(x + y + z - 4) + 2x\lambda = 0 \\ L'_y = 2(x + y + z - 4) + 2y\lambda = 0 \\ L'_z = 2(x + y + z - 4) + 8z\lambda = 0 \\ L'_\lambda = x^2 + y^2 + 4z^2 - 1 = 0 \end{cases} \quad 4分$$

解之得 $x = y = \pm 2/3, z = \pm 1/6$. 6分

$d(-2/3, -2/3, -1/6) = \frac{11}{2\sqrt{3}}$ 最大值, $d(2/3, 2/3, 1/6) = \frac{5}{2\sqrt{3}}$ 最小值. 8分

五 (共10分).

$$\text{设 } A = \int_0^1 dx \int_{x^2}^1 f(x, y) dy, f(x, y) = \frac{xy}{\sqrt{1+y^3}} - A. \quad 2\text{分}$$

$$\text{两边积分得: } A = \int_0^1 dx \int_{x^2}^1 \frac{xy}{\sqrt{1+y^3}} dy - \int_0^1 dx \int_{x^2}^1 A dy. \quad 4\text{分}$$

$$\int_0^1 dx \int_{x^2}^1 \frac{xy}{\sqrt{1+y^3}} dy = \int_0^1 dy \int_0^{\sqrt{y}} \frac{xy}{\sqrt{1+y^3}} dx = \int_0^1 \frac{y}{\sqrt{1+y^3}} \frac{1}{2} x^2 \Big|_0^{\sqrt{y}} dy = \frac{\sqrt{2}-1}{3}.$$

$$\int_0^1 dx \int_{x^2}^1 A dy = \frac{2}{3} A. \quad 8\text{分}$$

$$\text{故 } A = \frac{\sqrt{2}-1}{5}, f(x, y) = \frac{xy}{\sqrt{1+y^3}} - \frac{\sqrt{2}-1}{5}. \quad 10\text{分}$$

六 (共8分). $P = \sqrt{1+x} + y^2 e^{xy^2}, Q = 2xy e^{xy^2}$ 2分

$$Q'_x = 2y e^{xy^2} + 2xy e^{xy^2} y^2, P'_y = 2y e^{xy^2} + y^2 e^{xy^2} 2xy, Q'_x = P'_y. \text{ 积分与路径无关. } 6\text{分}$$

$$\int_L = \int_{(0,0)}^{(2,0)} = \int_0^2 \sqrt{1+x} dx = 2\sqrt{3} - \frac{2}{3}. \quad 8\text{分}$$

七 (共10分). 设 Σ_1 为平面 $z = 1$ 含在球内的部分, 取上侧.

$$P = x^3(1+y^3), Q = y^3(1+x^3), R = (z^3 + x^3 y^3). \quad 1\text{分}$$

由高斯公式

$$\begin{aligned} \oint_{\Sigma+(-\Sigma_1)} &= \iiint_{\Omega} [P'_x + Q'_y + R'_z] dv \\ &= \iiint_{\Omega} [3(x^2 + y^2 + z^2) + 3x^2 y^2 (x + y)] dv = 3 \iiint_{\Omega} (x^2 + y^2 + z^2) dv. \end{aligned}$$

$$(\text{由对称性, } \iiint_{\Omega} 3x^2 y^2 (x + y) dv = 0) \quad 5\text{分}$$

$$\begin{aligned} 3 \iiint_{\Omega} (x^2 + y^2 + z^2) dv &= 3 \int_1^{\sqrt{2}} dz \iint_{D_z} (x^2 + y^2 + z^2) dx dy \\ &= 3 \int_1^{\sqrt{2}} dz \int_0^{2\pi} d\theta \int_0^{\sqrt{2-z^2}} \rho(\rho^2 + z^2) d\rho = \frac{3\pi}{10} (16\sqrt{2} - 19). \end{aligned} \quad 7\text{分}$$

$$\Sigma_1 : z = 1, D_{xy} : x^2 + y^2 \leq 1. \iint_{\Sigma_1} = \iint_{D_{xy}} (1 + x^3 y^3) dx dy = \iint_{D_{xy}} 1 dx dy = \pi. \quad 9\text{分}$$

$$\text{故 } \iint_{\Sigma} = \frac{3\pi}{10} (16\sqrt{2} - 19) + \pi. \quad 10\text{分}$$

八 (共8分).

$$e^{2x} = \sum_{n=0}^{\infty} \frac{1}{n!} (2x)^n = \sum_{n=0}^{\infty} \frac{2^n}{n!} x^n, R_1 = +\infty. \quad 2\text{分}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, R_2 = 1. \quad 4\text{分}$$

$$\text{故 } f(x) = \left(\sum_{n=0}^{\infty} \frac{2^n}{n!} x^n \right) \left(\sum_{n=0}^{\infty} x^n \right) = \sum_{n=0}^{\infty} c_n x^n,$$

$$\text{其中 } c_n = \sum_{k=0}^n \frac{2^k}{k!}. \quad 6\text{分}$$

当 $x = \pm 1$ 时, $|c_n x^n| = c_n \not\rightarrow 0$, 故 $\sum_{n=0}^{\infty} c_n x^n$ 收敛域为 $(-1, 1)$.

$$f^{(10)}(0) = 10!c_{10} = 10! \sum_{k=0}^{10} \frac{2^k}{k!}. \quad 8分$$

九 (共10分).

$$(1) \Sigma: z = (x^2 + y^2 + z^2)^2. \quad 3分$$

(2)注意: $0 \leq z \leq 1$ ($\because z \leq z^4$). 形心坐标为 $(0, 0, \bar{z})$.

$$\bar{z} = \frac{\iiint_{\Omega} z dv}{\iiint_{\Omega} dv}$$

$$D_z: x^2 + y^2 \leq \sqrt{z} - z^2.$$

$$\begin{aligned} \iiint_{\Omega} dv &= \int_0^1 dz \iint_{D_z} dxdy = \int_0^1 dz \int_0^{2\pi} d\theta \int_0^{\sqrt{\sqrt{z}-z^2}} \rho d\rho \\ &= \pi \int_0^1 (\sqrt{z} - z^2) dz = \frac{\pi}{3}. \end{aligned} \quad 6分$$

$$\iiint_{\Omega} z dv = \int_0^1 z dz \iint_{D_z} dxdy = \int_0^1 z dz \int_0^{2\pi} d\theta \int_0^{\sqrt{\sqrt{z}-z^2}} \rho d\rho = \frac{3\pi}{20}. \quad 9分$$

$$\bar{z} = \frac{9}{20}. \text{ 形心: } (0, 0, 9/20). \quad 10分$$

十 (共10分).

(1) $p = 5/4$, 绝对收敛.

$$\lim_{n \rightarrow \infty} \frac{|e^{(-1)^n n^{-5/4}} - 1|}{n^{-5/4}} = 1. \quad 3分$$

(2) $p = 3/4$, 条件收敛.

由泰勒公式, $e^{(-1)^n n^{-3/4}} - 1 = (-1)^n n^{-3/4} + \frac{1}{2} n^{-3/2} + o(n^{-3/2})$.

$\sum (-1)^n n^{-3/4}$ 条件收敛, $\sum [\frac{1}{2} n^{-3/2} + o(n^{-3/2})]$ 绝对收敛. 故原级数条件收敛. 6分

(3) $p = 1/4$, 发散.

$$\begin{aligned} \text{由泰勒公式, } e^{(-1)^n n^{-1/4}} - 1 &= (-1)^n n^{-1/4} + \frac{1}{2} n^{-1/2} + \frac{1}{3!} (-1)^n n^{-3/4} + \frac{1}{4!} n^{-1} + \\ &\frac{1}{5!} (-1)^n n^{-5/4} + o(n^{-5/4}). \end{aligned} \quad 8分$$

$\sum (-1)^n n^{-1/4}$, $\sum \frac{1}{3!} (-1)^n n^{-3/4}$ 条件收敛, $\sum [\frac{1}{2} n^{-1/2} + \frac{1}{4!} n^{-1}]$ 发散, $\sum [\frac{1}{5!} (-1)^n n^{-5/4} + o(n^{-5/4})]$ 绝对收敛. 故原级数发散. 10分